Summary of Backpropagation-Free Training

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1 Experiment

Last week, I obtained the author's original code, which I utilized to evaluate various properties of the ULR method, including accuracy, training speed, and memory usage. First, the accuracy of the ULR method on the CIFAR-10 dataset matched the level reported in the paper [1]. The tests were conducted using the VGG-8 and ResNet-5 architectures. Table 1 compares the accuracy across different scenarios.

Table 1: Classification accuracies for different methods.

Training Strategy	**Model**	**Test Accuracy (%)**
Back-Propagation	ResNet 5 (Reported data)	64.4
	ResNet 5 (Primary data)	75.8
	VGG 8 (Reported data)	79.8
	VGG 8 (Primary data)	78.4
ULR	ResNet 5 (Reported data)	64.8
	ResNet 5 (Primary data)	67.5
	VGG 8 (Reported data)	77.3
	VGG 8 (Primary data)	76.1

Second, the memory requirements of ULR are typically tens of times higher than those of backpropagation under normal circumstances. In extreme cases, the memory demands of ULR may be equal to those of backpropagation, but they are never significantly lower than those of backpropagation. Table 2 illustrates the memory requirements for different training methods.

Table 2: Memory usage for different methods.

Training Strategy	**Model**	**Memory Usage (MB)**
Back-Propagation	ResNet 5	584
	VGG 8	596
ULR	ResNet 5 (Normal)	20076
	ResNet 5 (Extreme Case)	572
	VGG 8 (Normal)	15638
	VGG 8 (Extreme Case)	590

Third, as we assumed before, the training time of ULR is much longer than that of backpropagation. Table 3 shows the time required to train 100 epochs with different training methods.

Table 3: Training time for different methods.

Training Strategy	**Model**	**Training Time (h)**
Back-Propagation	ResNet 5	0.4
	VGG 8	0.37
ULR	ResNet 5	10.42
	VGG 8	14.72

2 Rethinking ULR

The memory inflation observed in the ULR method stems from its memory-for-time strategy, which allocates a large amount of memory to perform parallel computations for 100 repetitions in a single run. This approach aims to reduce training time by completing all repetitions simultaneously. However, even with this optimization, the training speed is still dozens of times slower than that of backpropagation. If the ULR method were to avoid this memory-for-time strategy, its memory requirements would be comparable to those of backpropagation. However, this would come at the expense of training thousands of times slower than backpropagation.

References

[1] J. Jiang, Z. Zhang, C. Xu, Z. Yu, and Y. Peng, "One forward is enough for neural network training via likelihood ratio method," *arXiv preprint arXiv:2305.08960*, 2023.